STRATEGIC CAPITAL PORTAL

REVENUE MODEL & GROWTH TRAJECTORY

Financial Projections

DOCUMENT TYPE	CLASSIFICATION	DATE
Investor Material	STRICTLY CONFIDENTIAL	March 9, 2026

MASTER PROJECT ROADMAP: FarmSense Strategy & Funding (V2.1)
The Single Source of Truth for Project Phasing, Grants, and Market Intelligence Consolidates: ROADMAP.md, FUNDING_PIPELINE.md, MILESTONES.md, 2 Strategy KIs, and high-stakes Grant Dossiers.

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1. Project Vision & Mission

FarmSense is the definitive sovereign water infrastructure for the 21st century. Its mission is to provides a legally recognized, cryptographically secure, and scientifically absolute **Digital Water Ledger** for global hydrological stability.

Core Strategic Mandates

The 1% Interpolation Physics: Establish high-resolution ground truth on 1% of a basin's footprint to mathematically interpolate 99% of the aquifer depletion via Kriging.

Deterministic Sovereignty: Replace stochastic "intuition" with auditable logic defensible in Water Court.

Dual-Use Resilience: Leverage commercial civilian AgTech as a self-sustaining layer for strategic inter-agency water intelligence.

2.1 Subdistrict 1 Hyper-Local Financial Dossier

This constitutes the authoritative statistical grounding for all market sizing and procurement.

Infrastructure Footprint: **1,280 active pivots** (~160,000 irrigated acres).

The Hydro-Economic Crisis: Aquifer depletion of 89,000 acre-feet/year. Groundwater extraction fees at **$500/acre-foot**.

Financial Savings: 52 AF average savings per pivot = **$26,000/year operational savings** per field.

TAM (Subdistrict 1 Bulk Rollout):

Hardware CapEx (One-time): $1,423,360 (at $1,112/PMT and $158.20/VFA scale).

SaaS ARR (Gold Truth): $7.6M/year at Enterprise rates ($30/acre).

Total Phase 1/2 Pilot Cost (2 Fields): **$12,450** (Estimated Single-Unit BOM).

Phase 2 Expansion (Full Truth): Estimated **$3,970,638** (total CAPEX for 18-node density).

2.1 The Revenue Model: The "Resolution Pop"

FarmSense captures market share via a tiered spatial fidelity model, driving conversion through high-resolution data visualization.

Tier	Grid Res	Cost	Focus	Behavioral Trigger
FREE	50m	$0	Gov / Compliance	"Macro-Truth" onboarding magnet.
BASIC	20m	$15/acre	Small Farmers	Initial Sentinel-2 processing.
PRO	10m	$22/acre	Commercial Ag	Broad-acre drone/sensor fusion.
ENTERPRISE	1m	$30/acre	Legal Fortress	**Gold Truth**: Sub-cm legal evidence.

The "FOMO" Trigger: When a user zooms in on a suspected anomaly (e.g., nozzle leak), the UI generates a high-contrast, blurred preview of the 1m grid with the overlay: *"High-Resolution Proof Available."* This converts technical curiosity into Immediate Enterprise upgrades.

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3. Phased Deployment Roadmap (2026–2030)

Phase	Timeline	Primary Objective	Geopolitical Goal
I: Pilot	Q1-Q2 2026	2-Field CSU Pilot	Establish "Rule-Compliant" digital subdistrict.
II: Regulatory Capture	Q3-Q4 2026	DWR Integration	Establish "Presumed Compliant" well status workflow.
III: State Standard	2027	Colorado Statewide	Primary tool for Colorado River Compact compliance.
IV: National Layer	2028	High Plains Aquifer	Standardize "Federal Water Credits" with USDA/USGS.
V: Sovereign Global	2029+	International G2G	Trans-boundary conflict mitigation (Nile/Indus).

3.4 The Seasonal "Rapid Deployment" Lifecycle

To maximize hardware lifecycle (10 years), FarmSense operates on a two-phase seasonal deployment model:

Post-Planting Insertion: Once spring planting is complete, crews drop high-value sensor sleds into the permanent HDPE SDR9 shells. Insertion, locking, and +5 PSI Nitrogen re-pressurization are completed in **under 15 minutes per field**.

Harvest Extraction: Prior to fall harvest, crews pull the C&C caps and extract the sensitive electronics for storage in the **RSS Sled Hospital**.

Winter Hibernation: Blanking plugs seal the permanent in-ground shells, while batteries undergo trickle-charging and diagnostics at the RSS hub,

preventing sub-zero degradation.

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4. Immediate Milestones & Deliverables

4.1 Beta Completion (Late Feb / Early March 2026)

CONFIDENTIAL — March

Software: Full Ingestion Pipeline + k-means Kriging Engine operational.

Frontend: Farmer Dashboard Beta (MapLibre GL JS) live on `brodiblanco.zo.computer`.

4.2 Beta Launch (CSU SLV Pilot - March 10, 2026)

Deployment: 2 Fields at CSU Research Center (Center, CO).

Hardware: 2 PMTs, 2 PFAs, 2 VFAs, 16-20 LRZ1/LRZ2 Nodes.

Goal: <5% Kriging Error validation for Water Court.

4.3 V1 Master Standard (Spring 2027)

Scale: Instrumenting all 1,280 SD1 fields + 25 DHUs.

Standard: Official integration into Colorado River Compact compliance.

Tech: AllianceChain PBFT Ledger fully active for P2P water trading.

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5.2 Mid-2026 Scaling (Requires Pilot Data)

NSF SBIR Phase I: $275K (Q2 2026)

CWCB Water Plan Grant: $500K–$2M (July 1, 2026)

BoR WaterSMART: $300K (July 31, 2026)

5.3 Global Impact Grants (Q3–Q4 2026)

Gates Foundation: COP30 COP30 Smallholder Advisory Mesh ($50.80 LRZ1 target).

World Food Prize: $250,000 nomination for food-security impact.

Earthshot Prize: £1M for climate/nature protection.

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6. Economic Logic: The “Resolution Pop”

FarmSense captures market share via a tiered spatial fidelity model, driving conversion through high-resolution data visualization.

Tier	Resolution	Cost	Focus	Target User
FREE	50m	$0	macro-Truth	Gov / Baseline Baseline Enrollment
BASIC	20m	$49/mo	Optimization	Small Family Farmers
PRO	10m	$199/mo	Daily Gain	Commercial Ag / 24hr VRI
ENTERPRISE	1m	Custom	**Legal Fortress**	Water Court / ESG / Trade

6.1 The “FOMO” UI Trigger

Resolution Pop: When a free/basic user zooms into a field anomaly, the 1m grid blurs and renders a “Locked” overlay.

Audited Proof: The UI showcases a snippet of the SAT angular movement or VFA deep-percolation anchor to prove the system has “Ground Truth” the user doesn’t.

Conversion: High-margin Enterprise upgrades occur precisely during high-stress hydrological events (Nozzle drops, irrigation surges).

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7. Geopolitical Dual-Use Positioning

7.1 Project OMNI (Strategic Intelligence)

FarmSense acts as a distributed “hydrological radar.” Funding from civilian AgTech maintains the cover for inter-agency water monitoring in contested regions (Trojan Horse Model).

Stealth: LPI/LPD LoRa Mesh emissions operate below the noise floor.

Resilience: JADC2 adapters translate soil moisture into tactical trafficability scores.

7.2 Global Philanthropy (Advisory Mesh)

For smallholders ($50.80 LRZ1), the “Advisory Mesh” provides deterministic SMS commands without requiring expensive pump actuation hardware, subsidized by domestic Enterprise tiers.

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8. Success Metrics (KPIs)

Kriging Accuracy: `< 5% Error` vs Ground Truth VFAs.

Mesh Uptime: `> 99.5%` under 10ft heavy canopy.

Water Savings: `20-30%` reductions documented in AllianceChain.

Regulatory Status: `100% Alignment` with DWR Rule 12.0 requirements.

Classification: Master Project Asset | Single Source of Truth | Approved 2026-03-07